

Concrete Case for Classification: Sentiment Analysis

Example #1: Movie Reviews

-  • Unbelievably disappointing
-  • Full of zany characters and richly applied satire, and some great plot twists
-  • This is the greatest screwball comedy ever filmed
-  • It was pathetic. The worst part about it was the boxing scenes.

Sentiment Analysis

Example #2: Product Reviews



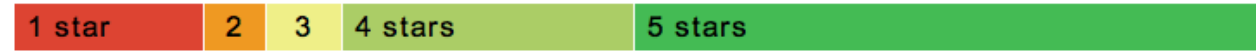
HP Officejet 6500A Plus e-All-in-One Color Ink-jet - Fax / copier / printer / scanner

\$89 online, \$100 nearby ★★★★★ 377 reviews

September 2010 - Printer - HP - Inkjet - Office - Copier - Color - Scanner - Fax - 250 sh

Reviews

Summary - Based on 377 reviews

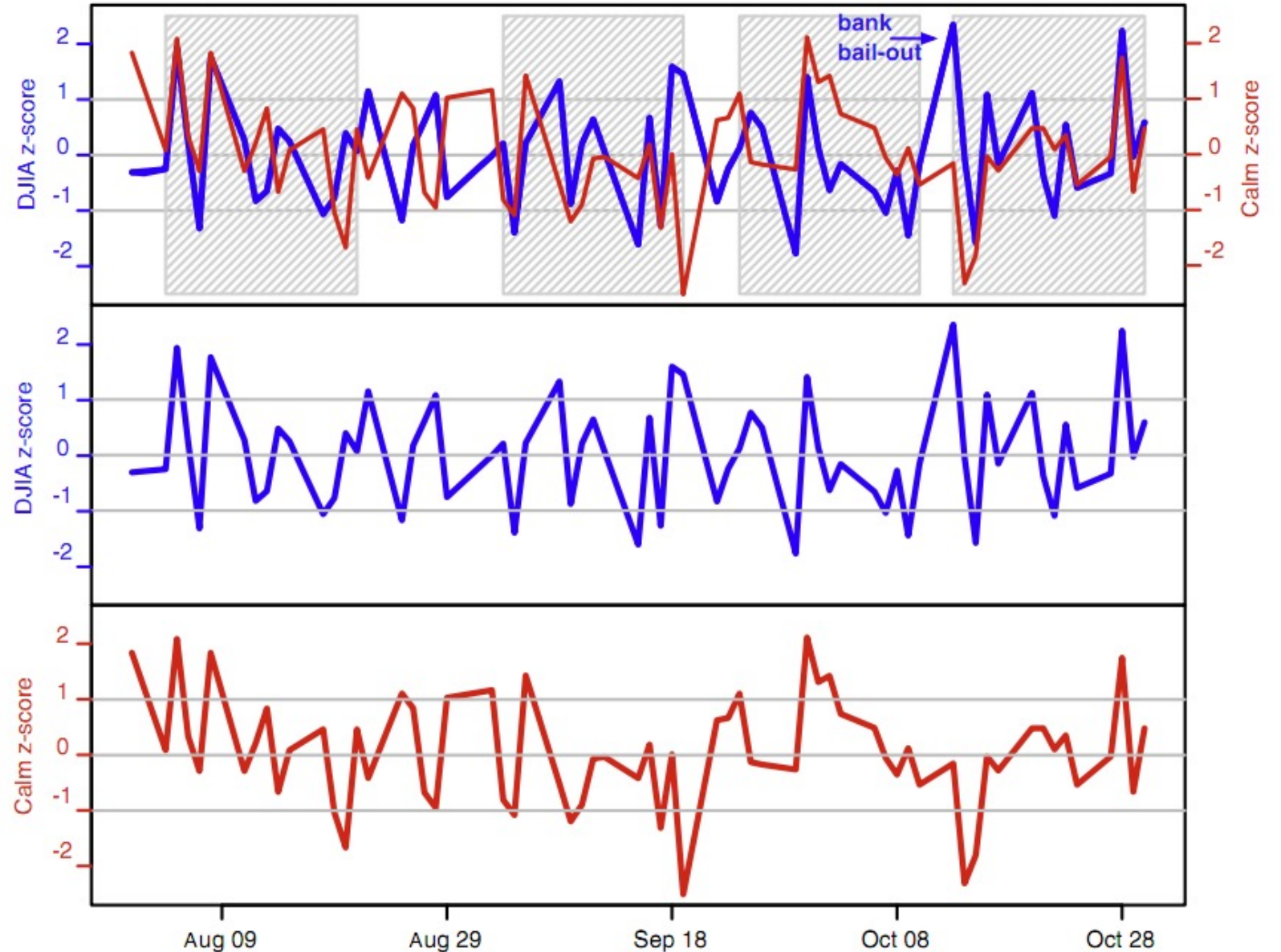


What people are saying

ease of use		"This was very easy to setup to four computers."
value		"Appreciate good quality at a fair price."
setup		"Overall pretty easy setup."
customer service		"I DO like honest tech support people."
size		"Pretty Paper weight."
mode		"Photos were fair on the high quality mode."
colors		"Full color prints came out with great quality."

- A Sentiment Analysis system called CALM applied to Twitter predicts the Dow Jones Industrial Average (DJIA) 3 days later
- Such algorithms are already used by hedge funds

Johan Bollen, Huina Mao, Xiaojun Zeng. 2011. [Twitter mood predicts the stock market](#), Journal of Computational Science 2:1, 1-8. 10.1016/j.jocs.2010.12.007.



Scherer Typology of Affective States

- **Emotion:** brief organically synchronized ... evaluation of a major event
 - *angry, sad, joyful, fearful, ashamed, proud, elated*
- **Mood:** diffuse non-caused low-intensity long-duration change in subjective feeling
 - *cheerful, gloomy, irritable, listless, depressed, buoyant*
- **Interpersonal stances:** affective stance toward another person in a specific interaction
 - *friendly, flirtatious, distant, cold, warm, supportive, contemptuous*
- **Attitudes:** enduring, affectively colored beliefs, dispositions towards objects or persons
 - *liking, loving, hating, valuing, desiring*
- **Personality traits:** stable personality dispositions and typical behavior tendencies
 - *nervous, anxious, reckless, morose, hostile, jealous*

Sentiment Analysis: Definition

- Sentiment analysis is the detection of **attitudes**
“enduring, affectively colored beliefs, dispositions towards objects or persons”
 1. **Holder (source)** of attitude
 2. **Target (aspect)** of attitude
 3. **Type** of attitude
 - From a set of types: *like, love, hate, value, desire*, etc.
 - Or (more commonly) simple weighted **polarity**: *positive, negative, neutral*, together with *strength*
 4. **Text** containing the attitude
 - Sentence or entire document

Sentiment Analysis

- Simplest task:
 - Is the attitude of this text positive or negative?
- More complex:
 - Rank the attitude of this text from 1 to 5
- Advanced:
 - Detect the target, source, or complex attitude types
- Sometimes not only complete sentences, but also their sub-parts receive a sentiment value
 - [The pizza was overall bad], but the toppings were fun! → overall negative

Sentiment Classification in Movie Reviews

- Is an IMDB movie review positive or negative?



when _star wars_ came out some twenty years ago
, the image of traveling throughout the stars has
become a commonplace image . [...]

when han solo goes light speed , the stars change
to bright lines , going towards the viewer in lines
that converge at an invisible point .

cool .



“ snake eyes ” is the most aggravating
kind of movie : the kind that shows so
much potential then becomes
unbelievably disappointing .

it's not just because this is a brian
depalma film , and since he's a great
director and one who's films are always
greeted with at least some fanfare .

Bo Pang, Lillian Lee, and Shivakumar Vaithyanathan. 2002. Thumbs up? Sentiment Classification using Machine Learning Techniques. EMNLP-2002, 79—86.

Bo Pang and Lillian Lee. 2004. A Sentimental Education: Sentiment Analysis Using Subjectivity Summarization Based on Minimum Cuts. ACL, 271-278

A Simple Classifier

- Log-linear or Naïve Bayes classifier
- Features:
 - The identity of the words
 - Possibly mark-up (e.g., hashtags in Twitter, headers in HTMLs)
- Features are often binary
 - Indicating whether a word appeared or did not appear in the document (bag of words)
 - Often works better for text classification than word frequency
 - The effect of frequency is non-linear

Error Analysis:

What makes reviews hard to classify?

- **Sarcasm:**

- Perfume review in *Perfumes: the Guide*:
 - “If you are reading this because it is your darling fragrance, please wear it at home exclusively, and tape the windows shut.”
- On *Automobile Steering Wheel Attachable Work Surface*:
 - “You wouldn’t believe how much more interesting my commute is now that I have something to do other than just stare out the window! I’m using it right now to post this review and I never”



- **Thwarted Expectations and Ordering Effects:**

- “This film should be brilliant. It sounds like a great plot, the actors are first grade, and the supporting cast is good as well, and Stallone is attempting to deliver a good performance. However, it can’t hold up.”

Negation in Sentiment Analysis

- One practice is to add NOT_ to every word between negation and following punctuation:

didn't like this movie , but I



didn't NOT_like NOT_this NOT_movie but I

Negation in Sentiment Analysis

- This is a very crude solution:
 - Explicit negation is only one way to reverse polarity
 - “He **failed** to convey the importance of his message”
 - Logical structure can be more complex
 - “I don’t think anyone could have done a better job”
 - Negation scope:
 - “I didn’t like the exposition in this otherwise excellent film”
- More recent approaches take the context (surrounding words) into account in the features
 - More about this later in the course